HW 2 [illegible] 113705013

C10S01Q13

13. • The data points (red dots) is distributed symmetrically to the blue line.
∴ The data is from normal population

C10S02Q20

20. $H_0 = \mu = 74$

$H_\alpha = \mu \neq 74$

t-test

$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{69.7 - 74}{16/\sqrt{25}} = -1.34 > -t_{24, 0.025} = -2.064$

∴ Non-reject H_0, we don't have enough evidence to proof the significant difference of Average CBF in healthy people and stroke patients

C10S02Q30

30. a. $H_0: \mu = 25$, $H_\alpha: \mu < 25$

$t = \frac{20.3 - 25}{.5/\sqrt{21}} = -4.307 < -t_{20, 0.05} = -1.725$

∴ Reject H_0, this is enough evidence to conclude that the true pretreatment mean is less than 25.

b. $26.6 \pm t_{20, 0.025} \times \frac{s}{\sqrt{n}} = 26.6 \pm 2.086 \times \frac{7.4}{\sqrt{21}}$

$= 26.6 \pm 3.368 \Rightarrow (23.23, 29.97)$

c. The self-esteem score increased from pretreatment to post-treatment and follow ups. This means that after going through the treatment, the self-esteem increased which means the treatment is successful.

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C10S02Q35

35. a. $H_0: \mu = 22$, $H_a: \mu > 22$

$$t = \frac{31 - 22}{6.2/\sqrt{17}} = 5.985 > t_{16, 0.05} = 1.746$$

∴ Reject H_0, there's enough evidence to proof that injected rats drink more water than uninjected rats

b. $31 \pm \underbrace{t_{16, 0.05}}_{1.746} \times \frac{6.2}{\sqrt{17}} = 31 \pm 2.625 \Rightarrow (28.37, 33.63)$

C10S03Q7

7. $\bar{x}_1 = \frac{12+3+8+5}{4} = 7$

$$s_1^2 = \frac{(12-7)^2 + (3-7)^2 + (8-7)^2 + (5-7)^2}{3} = 15.33$$

$\bar{x}_2 = \frac{14+7+7+9+6}{5} = 8.6$

$$s_2^2 = \frac{(14-8.6)^2 + (7-8.6)^2 + (7-8.6)^2 + (9-8.6)^2 + (6-8.6)^2}{4} = 10.3$$

$H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 \neq 0$

$df = n_1 + n_2 - 2 = 4 + 5 - 2 = 7$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{df} = \frac{3 \times 15.33 + 4 \times 10.3}{7} = 12.457$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{7 - 8.6}{\sqrt{12.457\left(\frac{1}{4} + \frac{1}{5}\right)}} = -0.675 > -t_{7, 0.025} = -2.365$$

R.R : $|t| > t_{7, 0.025} = 2.365$

p-value : $2 \times P[t < -0.675] \simeq 0.52 > 0.05$

∴ Non-reject H_0, our t-score lies between the boundaries of -2.365 to 2.365 and our p-value is way higher than 0.05.

C10S03Q8

8. $H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 \neq 0$

$df = 16 + 13 - 2 = 27$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{df} = \frac{15 \times 4.8 + 12 \times 5.9}{27} = 5.288$$

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$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{34.6 - 32.2}{\sqrt{5.288\left(\frac{1}{16}+\frac{1}{13}\right)}} = 2.795 > t_{27,\,0.025} = 2.052$$

R.R: $|t| > t_{27,\,0.025} = 2.052$

P-value $= 2 \times P(t > 2.795) \approx 0.0094 < 0.05$

$\therefore$ Reject H_0, t-score is outside of the boundaries and p-value is lower than 0.05

C10503012

12. a. $H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 \neq 0$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} = \frac{86.4 - 79.7}{\sqrt{\frac{8.95^2}{10}+\frac{10.7^2}{10}}} \approx 1.5188 < t_{18,\,0.025} = 2.101$$

$\therefore$ Non-reject H_0, t-score is in range, so there's no significant difference

b. $86.4 - 79.7 \pm \underbrace{t_{18,\,0.025}}_{2.101} \times \sqrt{\frac{8.95^2}{10}+\frac{10.7^2}{10}} \Rightarrow (-2.568,\ 15.968)$

c. From part a's conclusion and 95% CI, we can't tell if IPad is better or worse than traditional book because there's no significance difference

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C10503020

20. a. Above town $= x_1$, below town $= x_2$

$\bar{x}_1 = 5$

$s_1^2 = 0.025$

$\bar{x}_2 = 4.86$

$s_2^2 = 0.013$

$\frac{0.025}{0.013} \not> 3 \Rightarrow$ so, variances are equal

$$s_p^2 = \frac{4(0.025) + 4(0.013)}{8} = 0.019$$

① $H_0: \mu_1 - \mu_2 \leq 0$ vs $H_a: \mu_1 - \mu_2 > 0$

② $\alpha = 0.05$

③ Test statistic:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \sim t_8$$

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(4) $t^* = \dfrac{5 - 4.86}{\sqrt{0.019\left(\frac{1}{8}+\frac{1}{8}\right)}} \approx 1.606 < t_{8,0.05} = 1.860$

(5). R.R: $t > t_{8,0.05} = 1.860$

(6) Non-reject H_0, there is insufficient evidence to proof the inspector's claim.

b. 95% CI:

$$\bar{x}_1 - \bar{x}_2 \pm t_{8,0.025} \times \sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$$

$$= 5 - 4.86 \pm 2.306 \times \sqrt{0.019\left(\frac{1}{8}+\frac{1}{8}\right)} \Rightarrow (-0.061,\ 0.341)$$

C10 S04 Q20

20. Define $D_i = u_1 - u_2$

$D_i \sim N(\mu_D, \sigma_D^2)$, $i = 1, 2, \ldots, 20$

student	with	without	D_i	$D_i - \bar{D}_i$	$(D_i - \bar{D}_i)^2$
1	20	5	15	4.75	22.5625
2	24	9	15	4.75	22.5625
3	20	5	15	4.75	22.5625
4	18	9	9	-1.25	1.5625
5	22	6	16	6.75	33.0625
6	19	11	8	-2.25	5.0625
7	20	8	12	1.75	3.0625
8	19	11	8	-2.25	5.0625
9	17	7	10	-0.25	0.0625
10	21	9	12	1.75	3.0625
11	17	8	9	-1.25	1.5625
12	20	16	4	-6.25	39.0625
13	20	10	10	-0.25	0.0625
14	16	12	4	-6.25	39.0625
15	24	7	17	6.75	45.5625
16	22	9	13	2.75	7.5625
17	25	21	4	-6.25	39.0625
18	21	14	7	-3.25	10.5625
19	19	12	7	-3.25	10.5625
20	23	13	10	-0.25	0.0625
total			205		311.75

$\bar{D}_i = \dfrac{205}{20} = 10.25$

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$SD^2 = \dfrac{311.75}{19} = 16.407$, $SD = 4.050$

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① $H_0: \mu_D \le 0$; $H_A: \mu_D > 0$

② $\alpha = 0.05$

③ test statistic :

$$t = \frac{\bar{D}_i - D_0}{S_D/\sqrt{n}} \sim t_{19}$$

④ $t^* = \frac{10.25 - 0}{4.05/\sqrt{20}} = 11.32 > t_{19, 0.05} = 1.729$

⑤. R.R: $t > t_{19, 0.05} = 1.729$

⑥. Reject H_0, there's enough evidence to proof that average recall scorer is higher when imagery is used.

CIO S04 Q22

22. Define $\mu_1 - \mu_2 = D_i$

$D_i \sim N(\mu_D, \sigma_D^2)$, $i = 1, 2, \ldots, 6$

Location	Batter A	Batter B	D_i	$D_i - \bar{D}_i$	$(D_i - \bar{D}_i)^2$
1	0.135	0.129	0.006	0.016	0.000256
2	0.102	0.120	-0.018	-0.008	0.000064
3	0.098	0.112	-0.014	-0.004	0.000016
4	0.141	0.152	-0.011	-0.001	0.000001
5	0.131	0.135	-0.004	0.006	0.000036
6	0.144	0.163	-0.019	-0.009	0.000081
total			-0.06		0.000454

$\bar{D}_i = \frac{-0.06}{6} = -0.01$

$S_D^2 \approx 0.0000908$

$S_D \approx 0.00953$

a. ① $H_0: \mu_D = 0$ vs $H_A: \mu_D \neq 0$

② $\alpha = 0.05$

③ test statistic:

$$t = \frac{\bar{D}_i - D_0}{S_D/\sqrt{n}} \sim t_5$$

④ $t^* = \frac{-0.01}{0.00953/\sqrt{6}} = -2.571 = -t_{5, 0.025} = 2.571$

⑤ R.R: $t > 2.571$ or $t < -2.571$

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⑥ Non-reject H_0, not enough evidence to proof the difference between the average desire of orders

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b. 95% CI:

$$\bar{D}_i \pm t_{5,0.025} \times \frac{S_D}{\sqrt{n}} = -0.01 \pm 2.571 \times \frac{0.00753}{\sqrt{6}}$$

$$\Rightarrow (-0.02, 0)$$

C10S05Q12

12. a. completions:

$$\bar{x} = \frac{360}{16} = 22.5$$

$$s_1^2 = \frac{332}{15} \approx 22.133$$

$$s_1 \approx 4.70$$

Yards per pass:

$$\bar{x} = \frac{207.8}{16} = 12.99$$

$$s_2^2 = \frac{96.0775}{15} \approx 6.405$$

$$s_2 \approx 2.53$$

b. 95% CI: $\left(\frac{(n-1)s^2}{\chi^2_{15,\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{15,1-\alpha/2}}\right) = \left(\frac{15 \times 22.133}{27.4884}, \frac{15 \times 22.133}{6.26214}\right)$

($\alpha/2 = 0.025$, $1-\alpha/2 = 0.975$)

$$= (12.08, 53.02)$$

c. 95% CI: $\left(\frac{(n-1)s^2}{\chi^2_{15,\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{15,1-\alpha/2}}\right) = \left(\frac{15 \times 6.405}{27.884}, \frac{15 \times 6.405}{6.2644}\right)$

$$= (3.50, 15.34) \Rightarrow (1.87, 3.92)$$

d. ① $H_0: \sigma = 4$, $H_\alpha: \sigma \neq 4$

② $\alpha = 0.05$

③ Test statistic:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

④ $\chi^{2*} = \frac{15 \times 2.53}{4} \approx 6.0046 < 6.26214 = \chi^2_{15,0.975}$

⑤ R.R: $\chi^2 < 6.26214$ or $\chi^{2*} > 27.4884$

⑥ ∴ Reject H_0

C10S05Q13

13. a. $\bar{x} = \frac{353 + 351 + 351 + 355}{4} = 352.5$

$s^2 = 3.6667$

① $H_0: \sigma = .7$ vs $H_\alpha: \sigma \neq .7$

② $\alpha = 0.05$

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3. Test Statistic :

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1=3}$$

(4) $\chi^{2*} = \frac{3 \times 3.6667}{0.7^2} = 22.44 > \chi^2_{3,0.05} = 7.8147$

(5). R.R: $\chi^2 > 7.8147 = \chi^2_{3,0.05}$

(6). ∴ Reject H_0

b. 90% CI : $\left(\frac{(n-1)s^2}{\chi^2_{n-1,\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{n-1,1-\alpha/2}}\right) = \left(\frac{3 \times 3.6667}{7.8147}, \frac{3 \times 3.6667}{0.351846}\right)$

$\Downarrow$

$(1.4076, 31.2640)$

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17. a. No, because it shows such a big difference in variance

b. From part a, there's no evidence of a significance difference

Model:

$X_{1i} \sim N(\mu_1, \sigma_1^2)$, $i = 1, 2, \ldots, 15$

$X_{2j} \sim N(\mu_2, \sigma_2^2)$, $i = 1, 2, \ldots, 15$

a. $\dfrac{\text{Max}(s_1^2, s_2^2)}{\min(s_1^2, s_2^2)} = \dfrac{2.798095}{0.171143} > 3$ ∴ By rule of thumb,

b. ① $H_0: \mu_1 - \mu_2 = 0$, $H_a: \mu_1 - \mu_2 \neq 0$, ② $\alpha = 0.05$

③ Test statistic:

$$\frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim t_v$$

$$v = df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{(n_1-1)} + \frac{(s_2^2/n_2)^2}{(n_2-1)}} = 15$$

④ $t^* = \dfrac{3.73 - 4.8}{\sqrt{\frac{2.798095 + 0.171143}{15}}} = -2.41$

$-2.41 < -t_{15, 0.025} = -2.131$

∴ Reject H_0

⑤ R.R: $t < -2.131$ or $t > 2.131$

⑥ ∴ Reject H_0

9. D_i | 0.1 0.1 0 0.2 -0.1

$\bar{D}_i = 0.06$

$s_D^2 = 0.0132$

$s_D = 0.1147$

Model:

Define $D_i = u_1 - u_2$

$D_i \sim N(u_D, \sigma_D^2)$, $i = 1, 2, \ldots, 5$

① $H_0: u_D = 0$ vs $H_a: u_D \neq 0$

② $\alpha = 0.05$

③ Test statistic:

$$\frac{\bar{D}_i - D_0}{s_D / \sqrt{n}} \sim t_{(n-1)=4}$$

④ $t^* = \frac{0.06 - 0}{0.1147/\sqrt{5}} = 1.18 < t_{4, 0.025} = 2.776$

⑤ R.R: $t_i < -2.776$ or $t_i > 2.776$

⑥ ∴ Non-reject H_0

95% CI:

$\bar{D}_i \pm t_{4, 0.025} \times \frac{s_D}{\sqrt{n}} = 0.06 \pm 2.776 \times 0.051 \Rightarrow (-0.082, 0.202)$